

Regression and Sparsity

HW 5

Q1 "Probit Regression"

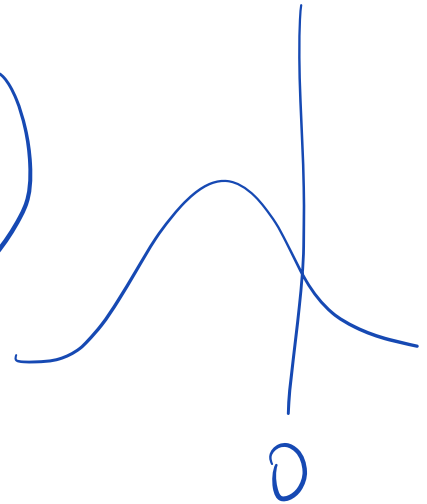
$$y_i \in \{0, 1\}$$

$$y_i \sim \text{Bern}(\Phi(\alpha + \beta X_i))$$

Data Augmentation

$$z_i \sim N(\alpha + \beta X_i, 1)$$

$$y_i = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z < 0 \end{cases}$$



Gibbs:

1. Sample $p(\alpha, \beta | z's)$ (Bayesian Regression)
2. Sample $p(z's | y, \alpha, \beta)$ (Truncated Normal)

Regression Models

$$Y_{n \times 1} = X_{n \times p} \beta_{p \times 1} + \epsilon$$

$$\epsilon \sim N(0, \sigma^2 I)$$

$P(\beta)$? What prior for regression coefficients?

$$y_n \sim N(X\beta, \sigma^2 I)$$

$P(\beta) \propto \text{const.}$ what is

$P(\beta | y, X)?$

$$\hat{\beta}_{MLE} = (X^T X)^{-1} X^T y$$

Linear
Regression

$$E[\hat{\beta}_{MLE}] = \beta$$

$$\text{Var}(\hat{\beta}_{MLE}) = \sigma^2 (X^T X)^{-1}$$

Frequentist
Property.

$$P(\beta | y, X) \sim N(\hat{\beta}_{MLE}, \sigma^2 (X^T X)^{-1})$$

Background

$$z \sim N_p(\mu, \Sigma)$$

$$P(z | \mu, \Sigma) \propto$$

$$|\Sigma|^{-1/2} \exp\left(-\frac{1}{2}(z-\mu)^T \Sigma^{-1}(z-\mu)\right)$$

Trace (sum of diagonal of a matrix)

Cyclical Invariance Property - (linear operator)

$$\text{tr}(ABC) = \text{tr}(BCA) = \text{tr}(CAB)$$

$$\begin{aligned}\text{tr}(A+B) &= \text{tr}(A) + \text{tr}(B) \\ k \cdot \text{tr}(A) &= \text{tr}(k \times A)\end{aligned}$$

Back to Regression

$$P(B|y, X) \propto P(y|X, B, \sigma^2) \propto$$

$$\exp\left(-\frac{1}{2} \underbrace{(y - XB)^T (\sigma^2 I)^{-1} (y - XB)}_{\text{scalar}}\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2} \left[\cancel{y^T y} - 2 \underbrace{B^T X^T y}_{\uparrow} + \underbrace{B^T X^T X B}_{\uparrow} \right]\right)$$

1x1 matrix

$$\propto \exp\left(-\frac{1}{2\sigma^2} \text{tr}\left(2 \underbrace{X^T y B^T}_{\uparrow} + \underbrace{(X^T X) B B^T}_{\uparrow}\right)\right)$$

If $X^T X$ is full rank

$$\propto \exp\left(-\frac{1}{2\sigma^2} \text{tr}\left((X^T X)^{-1} \underbrace{2(X^T X)^{-1} X^T y B^T}_{\uparrow} + B B^T\right)\right)$$

Complete
sq.

$\hat{\beta}_{MLE}$

$$\propto \exp\left(-\frac{1}{2\sigma^2} \text{tr}((X^T X)(\beta - \hat{\beta})^T (\beta - \hat{\beta}))\right)$$

$$\exp\left(\text{tr}\left(\underbrace{(\beta - \hat{\beta})}_{\text{mean}} \underbrace{\left(\frac{X^T X}{\sigma^2}\right)}_{\text{inverse cov}} (\beta - \hat{\beta})^T\right)\right)$$

$\Rightarrow$

$$P(\beta | y, X) \sim N(\hat{\beta}, \sigma^2 (X^T X)^{-1})$$

If $\beta \sim N_p(0, \sigma^2 I)$ (proper prior)
 what is the posterior?

$$P(B) \propto \exp\left(-\frac{1}{2\sigma^2} \text{tr}(B \Lambda^{-1} B^T)\right)$$

$$P(B|y, X) \propto L(B) P(B)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2} \text{tr}\left(\underbrace{X^T X B^T B}_{L(B)} - 2X^T y B + \underbrace{\Lambda^{-1} B^T B}_{P(B)}\right)\right)$$

$$\propto \exp\left(-\frac{1}{2\sigma^2} \text{tr}\left((X^T X + \Lambda^{-1})(B^T B + 2\underbrace{(X^T X + \Lambda^{-1})^{-1} X^T y}_{\hat{B}_{\text{ridge}}})\right)\right)$$

$$P(B|y, X) \sim N_p(\hat{B}_{\text{ridge}}, \sigma^2 (X^T X + \Lambda^{-1})^{-1})$$

Ridge Regression

$$\min_B \|y - XB\|_2^2 + \lambda \|B\|_2^2$$

$\uparrow$
 $L(B)$

$$\uparrow$$

$\log P(B)$

normal prior

R.R is equivalent
to maximizing the log posterior over B .

$$(X^T X + \lambda^{-1})^{-1} X^T y = \hat{\beta}_{\text{ridge}}$$

$$\underbrace{(X^T X + \lambda^{-1})^{-1} (X^T X)}_{\text{Shrinkage factor. (matrix)}} \underbrace{(X^T X)^{-1} X^T y}_{\hat{\beta}_{\text{MLE}} \text{ (OLS)}}$$

Identity

If X 's are orthonormal ($X^T X = I$)

and $\Lambda = \begin{pmatrix} 1 & 0 \\ 0 & \ddots \\ 0 & 0 & 1_p \end{pmatrix}$ (diag)

Then $(X^T X + \lambda^{-1})^{-1} (X^T X) = \text{diagonal with entries}$

$$\hat{\beta}_{\text{ridge}, i} = \frac{n}{n + \frac{1}{\lambda_i^2}} \hat{\beta}_{\text{MLE}, i}$$

$\frac{n}{n + 1/\lambda_j^2}$

- For large n , don't shrink much!
- For small n , or small λ , shrink a lot!

Bayesian (assuming $X^T X = I$)

$$E[B_j | \lambda_j, y, X] = \frac{n}{n + 1/\lambda_j^2} \hat{B}_{j, MLE}$$

$$= (1 - K_j) \hat{B}_{j, MLE}$$

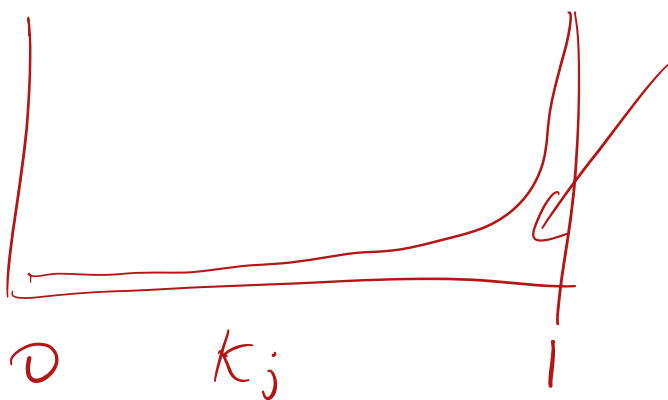
$$K_j = \frac{1}{1 + n\lambda_j} \quad (\text{Shrinkage Term})$$

Other Priors

Put a prior on λ_j 's and study
distribution of K_j .

$P(K_j)$

e.g.



Implies
High probability
of $B = 0$ (a priori)

Sparsity (The Bayesian Way)

- Lasso?

• “Spike and Slab”  Discrete Prior on Beta

• Horseshoe Priors  Cts Prior

- $R^2 D^2$ prior.

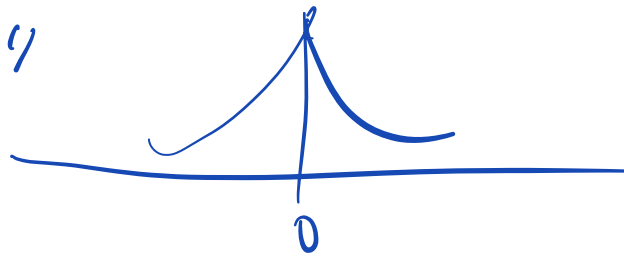
Lasso and Laplace Priors

Ridge: $\min_B \|y - XB\|_2^2 + \lambda \|B\|_2^2$

Lasso: $\min_B \|y - XB\|_2^2 + \lambda \|B\|_1$

$B_j \stackrel{\text{iid}}{\sim} \text{Laplace}, \quad P(B) \propto \exp(-\lambda \|B\|_1)$

"double exponential"



$$P(B|\dots) \propto L(B)P(B) \propto \exp(-\|y - XB\|_2^2 - \lambda \|B\|_1)$$

Only the MAP is likely to be
sparse.

Posterior Mean / Posterior samples will
likely not be sparse.

$p(y, b)$

Spike and Slab

$$B_j | \lambda_j, C, \epsilon \sim \lambda_j N(0, C^2) + (1 - \lambda_j) N(0, \epsilon^2)$$

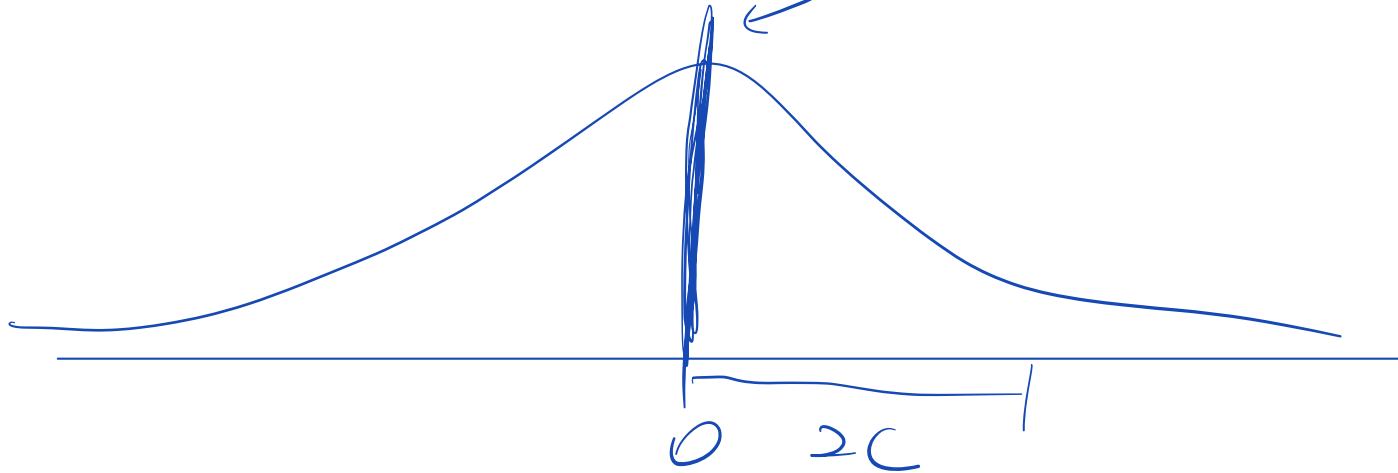
Assume $\epsilon \ll C$

Mixture of normals centered at 0 w/ variances

C & ϵ , and weights $\lambda_j / (1 - \lambda_j)$

$P(\lambda_j)$, $\lambda_j \sim \text{Bern}(p)$

$\epsilon \rightarrow 0$ / $(1 - p)$



$$B_j | \lambda_j, c \sim N(0, c^2 \lambda_j^2)$$

$$\lambda_j \in \{0, 1\}$$

$$\sim \text{Bern}(p)$$

Global
Shrinkage

(doesn't
depend
on i)

Local
Shrinkage

(does
depend
on j)

Global-Local

Shrinkage Priors

Problem: discrete parameters
means it can't be used w/ HMC.
Continuous analog?

Horseshoe

Sampling Variance

$$B_j \sim N(0, \sigma^2 \tau^2 x_j^2)$$

Global

Local

$$K_j = \frac{1}{1 + n \tau^2 x_j^2}$$

$$\tau^2 \rightarrow 0$$

$$B_{PM} \rightarrow \hat{B}_{MLE} \text{ (all Coefs)}$$

$$\tau^2 \rightarrow \infty$$

$$B_{PM} \rightarrow 0 \text{ (all coefs)}$$

$$B_{PM} = (1 - K_j) \hat{B}_{OLS}$$

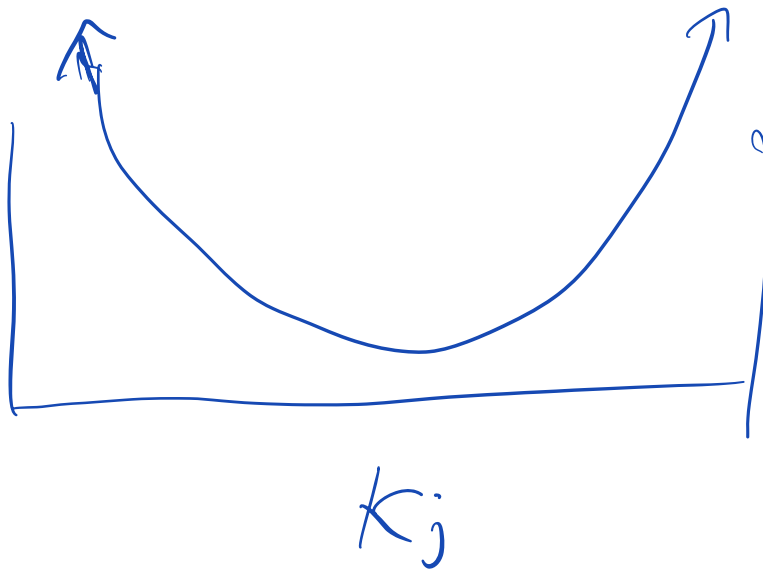
τ^2 controls overall sparsity.

Prior on λ_j (local shrinkage param)

$\lambda_j \stackrel{iid}{\sim} \text{Cauchy}^+ \text{ (half-cauchy)}$

$$\Rightarrow P(K_j | \tau^2) = \binom{1}{\sqrt{K_j} \sqrt{1-K_j}}$$

$$= K_j^{-1/2} (1-K_j)^{-1/2} = \text{Beta}(1/2, 1/2)$$

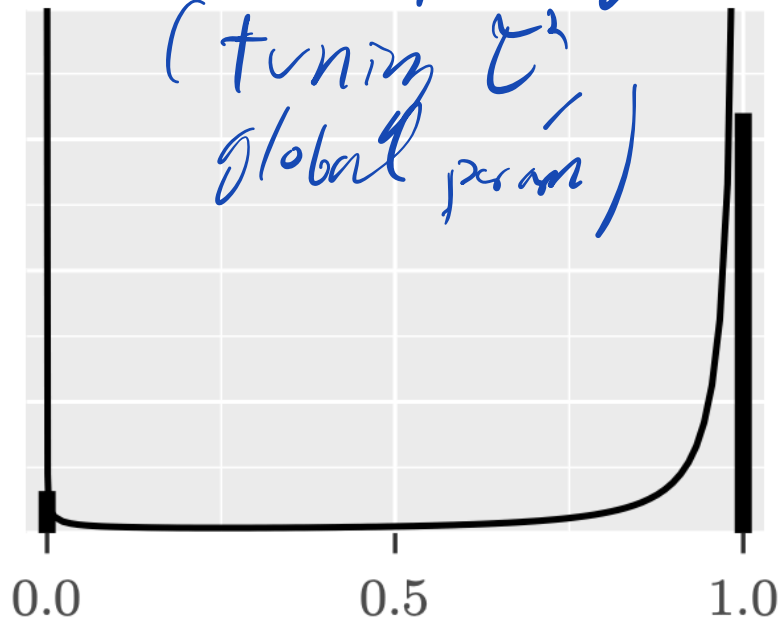
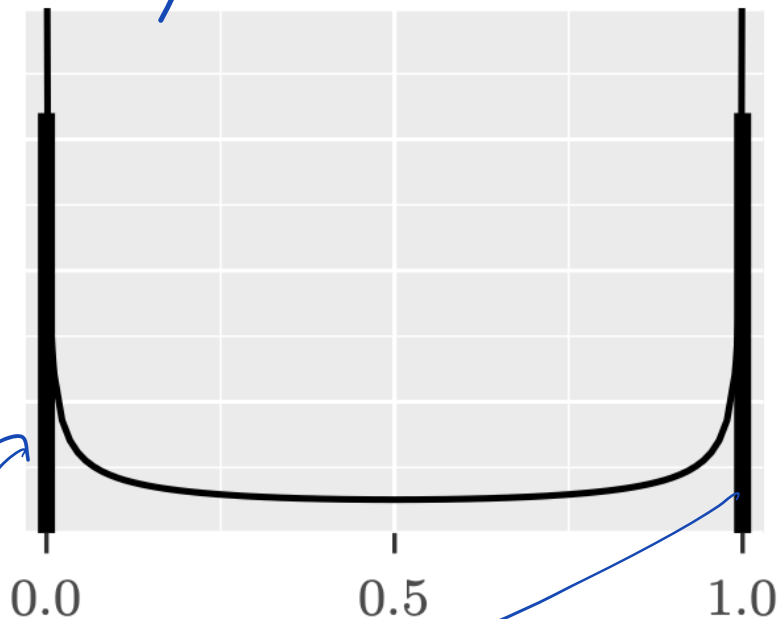


Horsehoe

(β_{MLE})

(shrink
to 0)

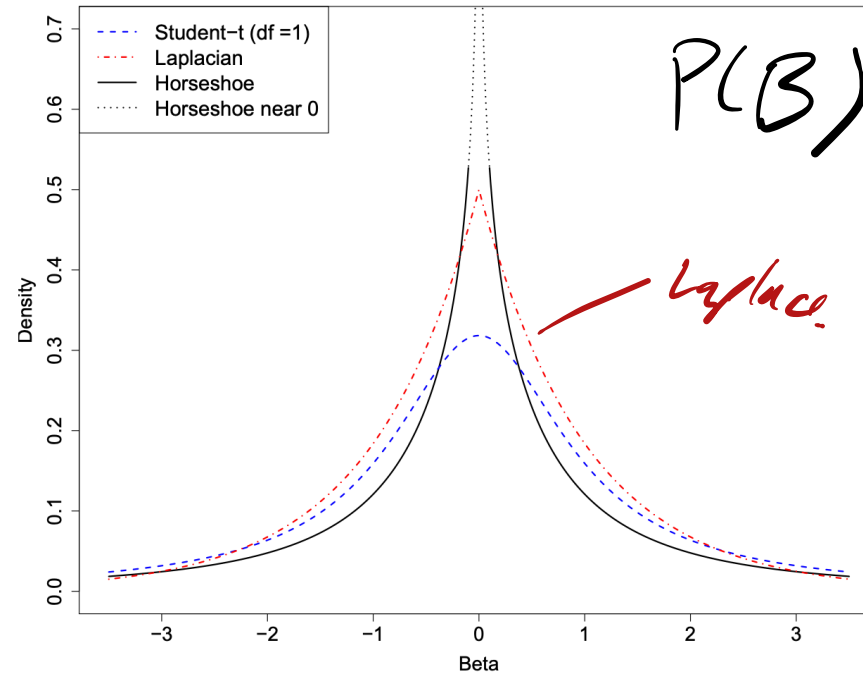
More Sparsity
(tuning τ^2
global prior)



Spike &
Slab.

Horseshoe is a cts approx.
to spike & slab.

Comparing Shrinkage Priors



Shrinkage Plots

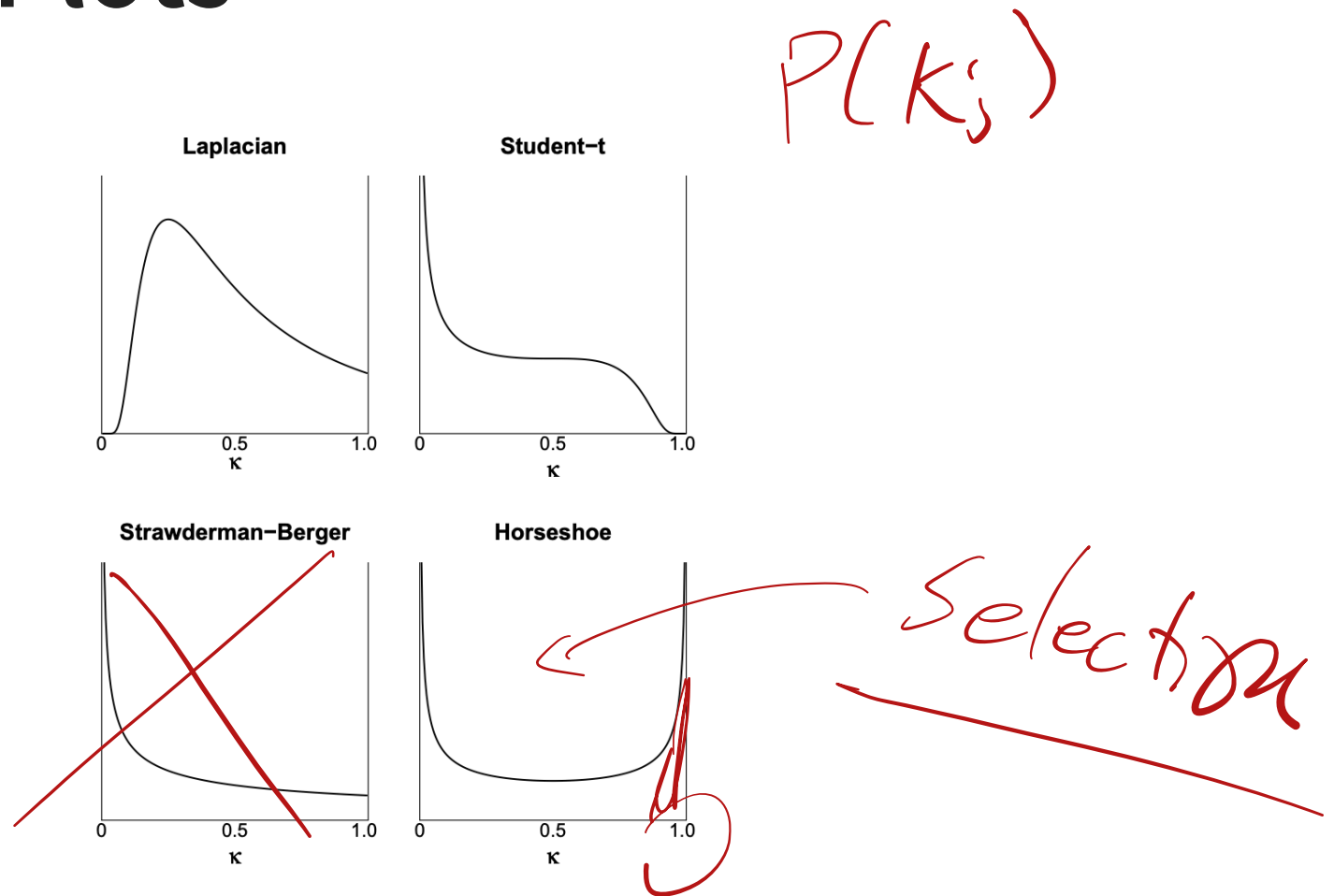


Figure 2: Densities for the shrinkage weights $\kappa_i \in [0, 1]$. $\kappa_i = 0$ means no shrinkage and $\kappa_i = 1$ means total shrinkage to zero.

Finnish Horseshoe

- Original horseshoe favors 0 or $\hat{\beta}$ as solutions
- When coefficients are poorly identified, it's helpful to add additional regulation for the largest coefficients
- The *regularized* horseshoe shrinks all coefficients by at least a small amount

Sparsity information and regularization in the horseshoe and other shrinkage priors

Regularized Horseshoe

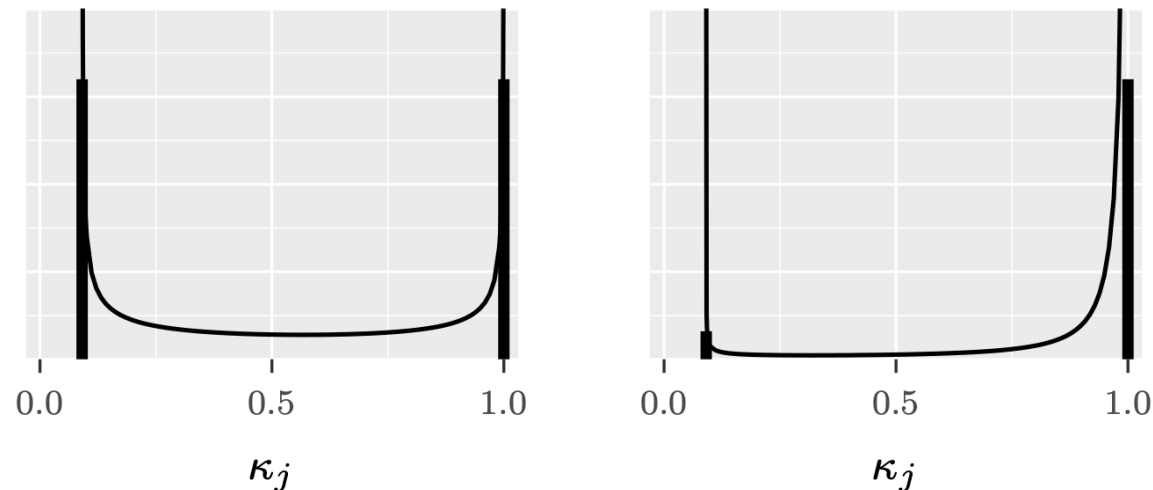


Fig 2: Shrinkage profiles as in Figure 1 but now for the regularized horseshoe (2.8) and spike-and-slab (2.7) with a finite slab width $c = 1$. In comparison to Figure 1, for both priors the first mode is shifted from $\kappa_j = 0$ to $\kappa_j = 1/(1 + n\sigma^{-2}s_j^2c^2)$ (for the plots we have selected $n\sigma^{-2}s_j^2 = 10$).

Global shrinkage parameter

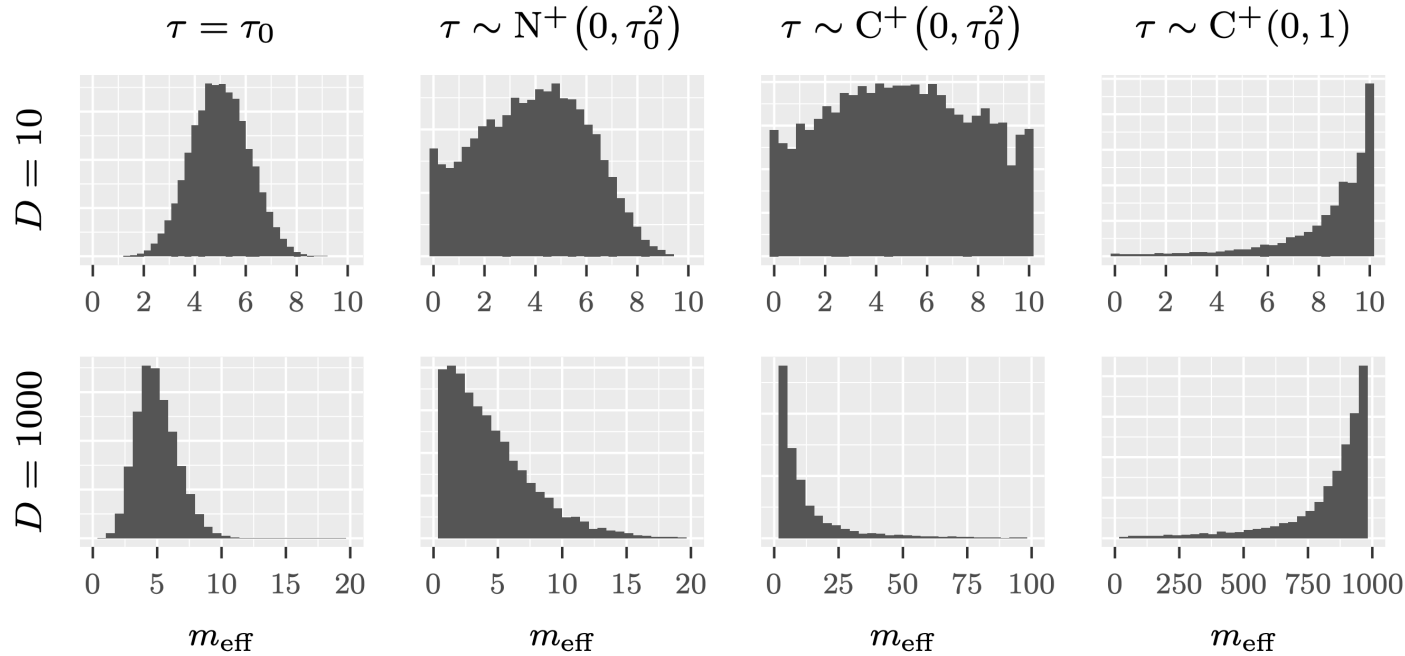


Fig 3: Histograms of prior draws for m_{eff} (effective number of nonzero regression coefficients, Eq. (3.7)) imposed by different prior choices for τ , when the total number of input variables is $D = 10$ and $D = 1000$. τ_0 is computed from formula (3.12) assuming $n = 100$ observations with $\sigma = 1$ and $p_0 = 5$ as the prior guess for the number of relevant variables. Note the varying scales on the horizontal axes in the bottom row plots.

R2-D2 Prior

- R2-induced Dirichlet decomposition (R2-D2)
- A typical approach: specify a p-dimensional prior on all regression coefficients
- Alternative: specify a prior on R^2 and then conditionally on β
 - For a fixed R^2 , β lies on an ellipsoid

Bayesian Regression Using a Prior on the Model Fit: The R2-D2 Shrinkage Prior

R2-D2

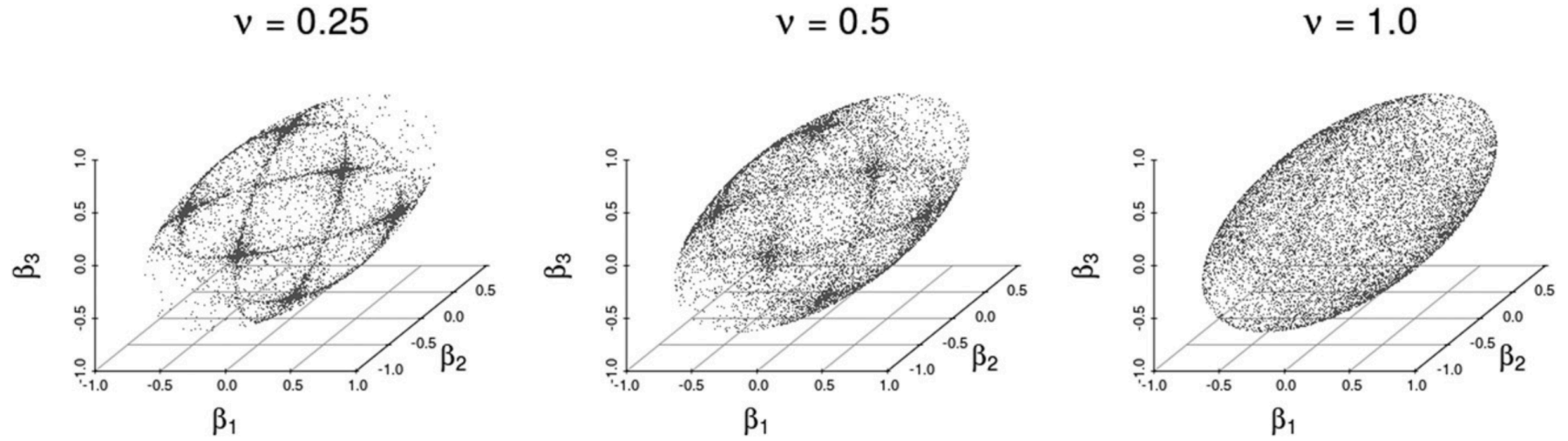


Figure 1. 10,000 samples from the prior of $\boldsymbol{\beta} \mid (R^2, \sigma^2)$ for different choices of ν .

Simulation

- $n = 50, p = 100$
- All coefficients are zero except for three of them
- Coefficients 2,4, and 6 have value 2
- Compare Gaussian, Laplace and Horseshoe priors using rstanarm.

```
1 library(rstanarm)
2 n <- 50
3 p <- 100
4 beta <- rep(0, p)
5 beta[1:3] <- 2
6 x <- matrix(rnorm(n*p), nrow=n)
7 x <- as.matrix(apply(x, 2, scale))
8 y <- rnorm(n=n, mean=x %*% beta, sd=1)
9
10 p0 <- 10
```

```
11 tau0 <- p0/(p-p0) * 1/sqrt(n)
12 fit_hs <- stan_glm(y~x, gaussian(), prior=hs(global_scale=tau0, slab_scale=
13
14 posterior <- as.matrix(fit_hs)
15 hs_dens <- mcmc_areas(posterior, pars=paste0("x", 1:10)) + xlim(c(-3, 3)) +
16 hs_dens
```

Gaussian Laplace fits

```
1 fit_gaus <- stan_glm(y~x, gaussian(), prior=normal(scale=1), data=tibble(x=
2 posterior_gaus <- as.matrix(fit_gaus)
3 gaus_dens <- mcmc_areas(posterior_gaus, pars=paste0("x", 1:10)) + xlim(c(-5
4
5 fit_lap <- stan_glm(y~x, gaussian(), prior=laplace(scale=1), data=tibble(x=
6 posterior_lap <- as.matrix(fit_lap)
7 lap_dens <- mcmc_areas(posterior_lap, pars=paste0("x", 1:10)) + xlim(c(-5,
```

R2-D2 Fit

```
1 library(brms)
2 options(mc.cores=4)
3 brm_fit <- brm(y ~ x, data=tibble(y=y, x=x),
4               prior=set_prior(R2D2(mean_R2 = 0.5, prec_R2 = 2,
5                                   cons_D2 = 0.1)),
6               family=gaussian())
```

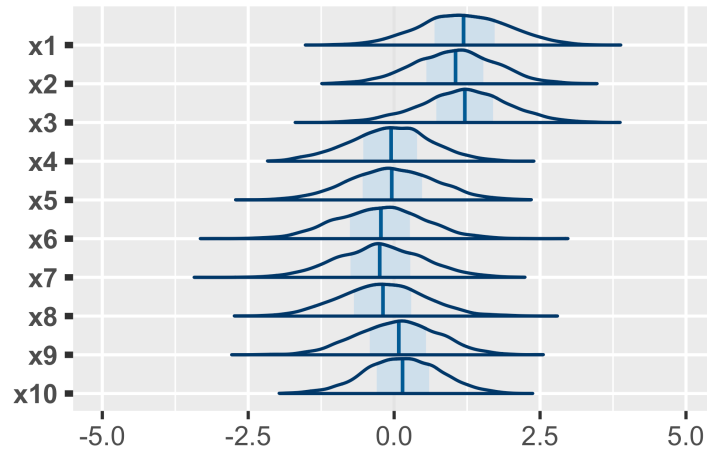
Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
using C compiler: 'Apple clang version 14.0.3 (clang-1403.0.22.14.1)'
using SDK: ''

```
clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -  
DNDEBUG -I"/Library/Frameworks/R.framework/Versions/4.3-  
arm64/Resources/library/Rcpp/include/" -  
I"/Library/Frameworks/R.framework/Versions/4.3-  
arm64/Resources/library/RcppEigen/include/" -  
I"/Library/Frameworks/R.framework/Versions/4.3-  
arm64/Resources/library/RcppEigen/include/unsupported" -  
I"/Library/Frameworks/R.framework/Versions/4.3-  
arm64/Resources/library/BH/include" -  
I"/Library/Frameworks/R.framework/Versions/4.3-  
arm64/Resources/library/StanHeaders/include/src/" -  
I"/Library/Frameworks/R.framework/Versions/4.3-
```

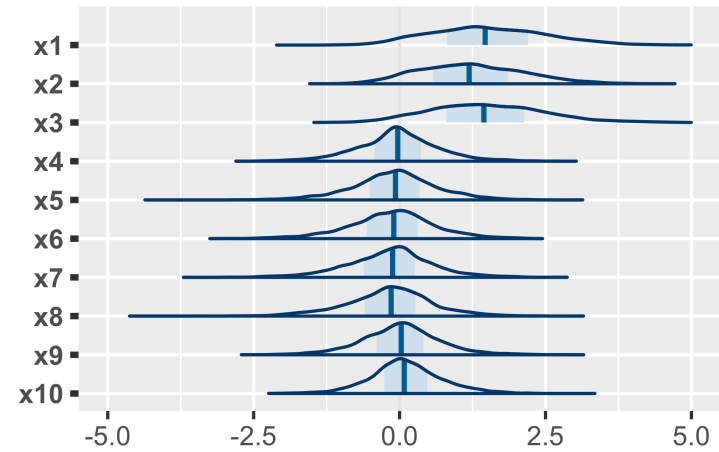
```
1 r2d2_dens <- mcmc_plot(brm_fit, type="areas", variable = paste0("b_x", 1:10)
```

Comparison Plots

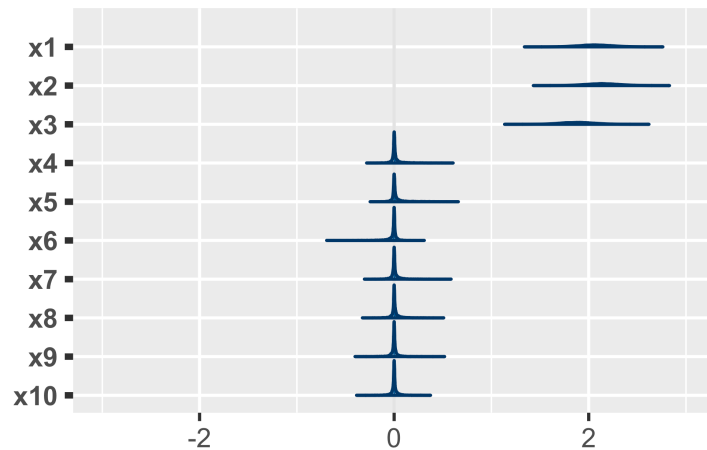
Gaussian



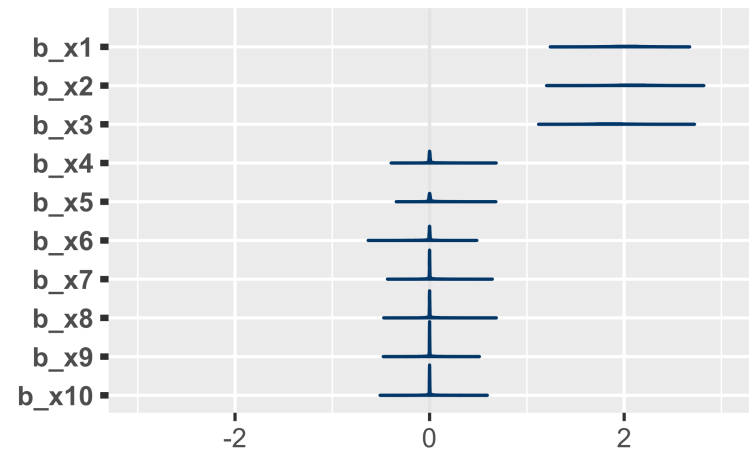
Laplace



Horseshoe



R2-D2



Loo Comparison

```
1 loo_hs <- loo(fit_hs)
2 loo_gaus <- loo(fit_gaus)
3 loo_lap <- loo(fit_lap)
4 loo_r2d2 <- loo(brm_fit)
5
6 loo_compare(loo_hs, loo_gaus, loo_lap, loo_r2d2)
```

	elpd_diff	se_diff
fit_gaus	0.0	0.0
fit_hs	-1.2	4.4
brm_fit	-3.6	3.3
fit_lap	-10.3	0.9

